Caius Brindescu, PhD

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About Me

Software Engineer with 15+ years of experience in building distributed systems and data infrastructure. Expert in back-end development, and cloud architecture, with a track record of leading cross-team initiatives and delivering measurable reliability and performance improvements. Skilled at debugging and resolving complex technical challenges across diverse technologies, throughout the entire tech stack. In my private life, I design and 3D-print practical solutions, read science-fiction and philosophy, and experiment with new technologies in my homelab.

Experience

Hightouch

Software Engineer – Distributed Systems (September 2026 – Present)

- Working on high-performance **realtime event processing**.

Etleap

Principal Engineer (November 2024 – August 2026)
Member of Technical Staff (September 2019 – November 2024)

- **Designed, and lead the implementation** of an in-VPC agent and control plane web app to allow customers to manage their private deployments without having to use Terraform.
- Led cross-team initiative spanning 3 development teams, **raising build reliability from 45% → 90%**.
- Owned and operated **AWS infrastructure** serving dozens of customers; developed zero-downtime upgrade process improving security posture.
- Worked with **Hadoop, Spark, Flink, AWS, MySQL, PostgreSQL, Redshift, Snowflake, Delta Lake** and **dbt**.
- Designing and maintaining **Terraform** templates for private customer deployments, as well internal module and appliances to simplify connectivity to customer's networks.
- Published **4 blogs** posts co-authored with senior AWS engineers.
- Migrated the build system to gradle from sbt to improve reliability. This involved adapting over **50 different modules** to use gradle.

Oregon State University

Graduate Assistant (August 2013 – June 2019)

During my time as a PhD student I worked as a Research and Teaching Assistant.

- Developed research tooling used by multiple projects and **published in peer-reviewed venues (6 conference, 2 journal)**
- Taught Software Engineering, Software Testing, Web Development, Operating Systems and Empirical Software Engineering, teaching classes of over 100 students.
- Instructor of Record for Software Engineering and Web Development classes (100+ students), redesigning course curriculum to improve student engagement and outcomes.

Etleap

Software Engineering Intern (June – September 2018)

- Worked on developing and maintaining the back-end code, including REST APIs, third party integrations or job orchestration.
- Developed 3 proof-of-concept integrations with external services.

University of Illinois at Urbana-Champaign
Graduate Teaching Assistant (August 2012 – August 2013)

Worked a teaching assistant for Software Engineering 1 and 2, under the guidance of Ralph Johnson.

Movidius (acquired by Intel)
Software Engineer (March – July 2012)

Coordinated a team of 4 developers to integrate an existing compiler, debugger and virtual board simulator into the Eclipse IDE.

Education

PhD in Computer Science
Oregon State University, Corvallis OR, 2020

Research in software engineering and developer productivity; published 8 peer-reviewed papers, 6 in conference proceedings and 2 in journals.

BS in Computer Science
Universitatea Politehnica "Timișoara", Timișoara, Romania, 2011

Course work included Software Engineering, Computer Architecture, Distributed Systems and Compilers.

Skills

Programming Languages
- **Core:** Java (15+ years).
- **Proficient:** Scala, Python, Bash, R.
- **Working knowledge:** TypeScript, C, C++, JavaScript, Go, Smalltalk

Frameworks and technologies
- **Data & Streaming**: Hadoop, Spark, Flink
- **Databases and SQL**: MySQL, PostgreSQL, Redshift, Snowflake, Databricks DeltaLake, dbt
- **Infra & DevOps**: AWS (certified), Terraform, Ansible, Kubernetes, Bash
- **Web/Apps**: Play Framework, Spring Boot, Flask

Leadership and Communication
- Authored peer-reviewed papers, technical blogs (including **4 AWS-published blog posts**), and internal documentation.
- Led cross-team initiatives (build systems, database migrations, Terraform modules design and maintenance, designed and delivered new greenfield products).
- Managed enterprise customer trials and on-boarding onto the platform.
- Directed research projects and mentored graduate and undergraduate students.
- Scoped and delivered new product features from design to production.